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$$\int \frac{7dx}{x^2+8x+18}$$
$$= \int \frac{7dx}{(x+4)^2+2}$$
$$= \int \frac{7dx}{2\left(\frac{(x+4)^2}{2}+1\right)}$$
$$= \frac{7}{2}\int \frac{dx}{\left(\left(\frac{x+4}{\sqrt{2}}\right)^2+1\right)}$$
$$u = \frac{x+4}{\sqrt{2}}$$
$$du = \frac{\sqrt{2}}{2}dx$$
$$\frac{2du}{\sqrt{2}} = \frac{2\sqrt{2}du}{2} = dx$$
$$\Rightarrow \frac{7\cdot 2\sqrt{2}}{2\cdot 2}\int \frac{1}{u^2+1}du$$
$$= \frac{7\sqrt{2}}{2}\,\mathrm{Bgtan}\left(\frac{x+4}{\sqrt{2}}\right)+C$$

References